FPGA circuit simplification
100nF = 0.1uF

04/28/2024
I think it's better that the address decoding fpga (FPGA 0) handles the bus stuff, since it will require any form of bus steering (for not scattered drams, at least)

I will have to use a clever arrangement of transceivers to make this all work between 2 144 pin fpgas

todo: update all the modified layouts 04/28/2024 THIS HASN'T BEEN DONE YET

other major changes to fpga circuit:
- spi rom OE is always 0 (correct mode and there's no jumper to change that
- only 1 schpe jumper is required for switching between turning off the fpga and normal operation (to make rom flash problems easier to recover from)
- fixed some mistakes in the schematic (the schematic guide says to do this, to make inventory management easier, the only SMD capacitors I use are 0.1uF and 1uF

FPGA 0

- address decoding
- cpu control
- everything

FPGA 1

- bus steering, interrupts and dma

POWER DELIVERY AND SYSTEM RESET

jumpers for every possible config combination

U70 74LS04

RESET: 0= dont reset, 1=reset

RESET: 1= dont reset, 0=reset.

reset active low
1 = reset
0 = don't reset

RESET

buffered just because

SYSTEM_RESET

FPGA_RESET_BLITTER

jumping pins 3 and 4 makes this a 0 instead of a 1

maybe a 1k resistor will make the power status led be not so blindingly bright

J25

J26 ATX-24

PS_ON#

PWR_OK

LED

J27 Conn_01x02

R41 R 1k

1k resistor = dim led
470ohm resistor = bright ed led

U52 74LS32 U70 74LS04 U45E 74LS08

system clock

Reset circuit

sometimes one is more

the ic809 works perfectly with lse n0 fpgas

U26 LM809

R30 R 100k

try it before putting this resistor on

cpu voltage selection

switch between the 5 volt rail or the regulated cpu rail

3.3 volts or 5.45 volts

enable, disable or disconnect WTWD pin

R27 10k

R28 10k

cpu option selection shift registers

make it easier for the user to configure cpu type

there was going to be stuff here to automatically configure cpu type but there just weren't enough available fpga pins for it so nevermind

the second version with all the issues from version 1.0 fixed

V2

fixed the sram addressing and used 72x instead of 80x for sram

let the fpgas take over as many tasks as possible this time.
Don't use discrete logic chips without a good reason.

control the address transceivers via fpga this time

make the fpgas control more of the system this time

(for 'for real' ABL (the one before this worked, making this a second version or something)
features:
-16 bit ram bus
-2 fpga chips, one ram, sram and also dram. dram has a 64x bus bank swapping system to make configuace 32 bit reads faster
-2 latch fpgas, 32 bit to 16 bit ram bus steering needs a lot of pins (80+)
-remake the second fpga also do dma and interrupts for ms-dos compatibility
-remake the code to enable a spi rom based bios is still in development, so a jmecatel rom is still required

Coding notes:
Flint 'for_real_ABL4.kicad_sch

Title: for_real_ABL4.kicad_sch

Size: A0 | Date: | Rev:
KiCad 8.0.4, 8.0.5 | Id: 1/1